

# SAI VIKRANTH KANURU

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## Career Objective

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B.Tech Electronics and Communication Engineering student building Generative AI and agentic systems, particularly Retrieval-Augmented Generation (RAG) applications. Passionate about solving problems end-to-end and developing production-grade AI solutions. Seeking an internship opportunity to contribute to real-world AI projects and collaborate with engineering teams.

## Technical Skills

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**Programming:** Python, C, SQL (MySQL, PostgreSQL), HTML, CSS

**Frameworks & Libraries:** LangChain, LangGraph, HuggingFace, FastAPI, Streamlit, Playwright, FastMCP

**Databases & Retrieval:** PostgreSQL, MySQL, Redis, Qdrant, FAISS, Chroma

**AI/ML Concepts:** LLMs, Embeddings, RAG, Prompt Engineering, Speech-to-Text (STT), Text-to-Speech (TTS), WebRTC

**Tools:** Docker, LangSmith, GitHub

## Projects

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- **Multi-Threaded WebRTC Voice Agent**
  - Built sub-100ms voice assistant using LiveKit WebRTC, Silero VAD, Deepgram STT, LLaMA 3.3, and Edge TTS.
  - Added voice-controlled browser automation with LangGraph workflows for website interaction and app triggers.
  - Implemented PostgreSQL memory and Redis caching for preference learning and 2ms API response optimization.
  - Created real-time 3D holographic UI using Three.js with WebRTC data synchronization.
- **Conversational Text2SQL Assistant & Obsidian Console**
  - Built natural language SQL assistant with intelligent routing and self-healing query execution.
  - Added SQL security layers with destructive command blocking and sensitive data masking.
  - Optimized Redis caching, connection pooling, and LangSmith tracing for query monitoring.
  - Developed console with live statistics, exports, and MySQL-to-PostgreSQL dialect conversion.
- **Multi-Source RAG AI Assistant**
  - Built document assistant supporting PDFs, websites, text files, and scanned documents using Llama-Parse OCR.
  - Implemented hybrid retrieval using semantic search, keyword matching, embeddings, and neural reranking.
  - Designed LangGraph workflows with PostgreSQL memory and persistent conversation history.
  - Built telemetry dashboards with retrieval metrics, audit trails, and LangSmith integration.

## Education

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### Bachelor of Technology – Electronics and Communication Engineering

MVGR College of Engineering, Vizianagaram

Expected May 2028

CGPA: 7.2

### Intermediate – Science (MPC)

Narayana Junior College, Vizianagaram

May 2024

CGPA: 8.72

### Class 10 – SSC

Fort City School, Vizianagaram

March 2022

CGPA: 8.58

## Certifications

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- Python Essentials 1 & 2 – Cisco Networking Academy
- Quantum Foundations – Andhra Pradesh Government

## Additional Information

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**Areas of Interest:** Generative AI, Information Retrieval, Agentic Systems, System Design, DSA

**Strengths:** Problem-solving, analytical thinking, system-oriented approach, rapid learning, adaptability, and teamwork.